Tactical choice in repeated protests

Elaine Yao*

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Abstract

Why do some protest movements escalate from peaceful demonstrations to violent confrontation while others never change tactics? I develop a dynamic model in which an organizer chooses between tactics that differ in baseline power and turnout sensitivity. In equilibrium, turnout is endogenously determined by public beliefs about regime strength. Failed protests shift those beliefs pessimistically, so the organizer balances immediate effectiveness against the informational consequences of failure. Tactical trajectories are governed by the relative comparison of tactics, as well as the magnitude of past failure. This generates three distinct regimes, which differ in both initial tactic choice and the degree of path dependence in the continuation. The framework yields testable predictions about tactical persistence and escalation, and offers a structural account of the observed correlation between nonviolent tactics and campaign success. On the technical side, the model also offers a resolution to equilibrium multiplicity in dynamic global games by having agents receive private cost rather than information signals.

*Assistant Professor, Washington University in St. Louis. E-mail: eyao@wustl.edu

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1 Introduction

When and why do protest movements change tactics? Some movements undergo striking transformations in their main mode of contestation. The African National Congress turned from civil disobedience to armed sabotage with the creation of Umkhonto we Sizwe; Arab Spring protesters shifted away from initial mass marches to violent clashes with security forces as initial demonstrations faltered in some cities; radical environmental movements have recently pivoted from large-scale marches to targeted acts of disruption and property destruction. Other movements never transform at all: Poland's Solidarity movement remained overwhelmingly nonviolent and mass-based throughout the 1980s, while a century earlier, January Uprising of 1863 took the form of armed insurrection from the start. What explains these divergent trajectories?

Existing theoretical accounts offer limited guidance, emphasizing instead how nascent protest movements grow by exposing latent sympathies, lowering barriers to participation and leading to cascades of information revelation (Kuran 1991; Lohmann 1994). Importantly, these accounts also neglect the question of *tactics* used in contentious politics, and how these tactics are strategically chosen before protesters choose whether or not to take to the streets. Furthermore, they only speak to the "optimistic" side of the information story. On the other side, as protest movements persist without having yet encountered success, the same mechanisms that once prompted their growth can turn against them: Rather than reveal the extent of public sympathies, each accumulated failure is suggestive of the weakness of the movement, and the strength of the regime or policy being contested.

This paper argues that tactical evolution – or constancy – is driven by how movements navigate this race against declining public opinion. The problem is not merely to choose the tactic that maximizes today's chances of success, but rather to choose tactics that are robust to the possibility of failure – and the accompanying loss of public confidence which constrains what is feasible tomorrow. Different tactics offer different tradeoffs along this frontier: some generate high expected power today but devastating informational consequences upon failure; others sacrifice immediate impact, but may better preserve the movement's future mobilization capacity. The paper shows that this tradeoff, together with the material characteristics of the tactics available, determines the trajectory of tactical choice, and whether or not this trajectory features persistence or learning-driven path dependence where the *type* of failure determines future tactics.

I develop this argument in a two-period model of repeated protest with endogenous turnout and tactical choice. An organizer selects between two tactics in each period: Peaceful mass tactics have high *turnout sensitivity*, and are therefore powerful with large crowds but weak without them, whereas violent or confrontational tactics have higher *baseline power* at low par-

ticipation, but lower turnout sensitivity, and hence a lower total ceiling. Turnout is generated by a modified global game in which citizens make individually optimal participation decisions given public beliefs about regime strength and privately realized costs. Failed protests lead citizens to update beliefs pessimistically, leading to lower expected future turnout. An organizer chooses tactics at the start of each period to maximize the probability of eventual success, understanding that today's tactical choice shapes both immediate prospects and the informational environment in which tomorrow's decisions are made. Hence, the stylization of repertoires of contention is deliberately chosen to match the channel through which public opinion enters the model.

At the heart of the model is a tension between immediate effectiveness and long-run viability: Tactics that maximize the probability of success today also generate failures that are maximally indicative of a strong regime, depressing expected turnout and therefore the efficacy of future action. A concrete illustration clarifies the logic. Consider a tactic whose power is entirely invariant to turnout – for instance, pre-positioned sabotage that requires no mass participation to execute. One might expect such a tactic to be attractive as an opening move, especially if its baseline power is high. In fact, a success-maximizing organizer *never* leads with it, no matter how powerful it is. The reason is that a turnout-insensitive tactic can be held in reserve for later use at no expected loss in effectiveness, since its power does not depend on public beliefs. The tactic whose power *does* depend on turnout – peaceful mass mobilization – is strongest precisely when beliefs are still optimistic, so the organizer uses it first and saves the turnout-invariant option for later. This result captures in its purest form the paper's central logic: what makes a tactic a good choice today is not only its power, but how its use influences the conditions under which future tactics will be deployed.

Building on this logic, I characterize the full set of equilibrium tactical paths using a linear parameterization. Three tactical regimes emerge, characterizing the full sweep of environments from those which unconditionally favor peace in both the first protest and the continuation – such as open urban settings with wide boulevards and public squares, limited access to weapons, and no insurgent traditions or strong nonviolent convictions – to environments in which the path of tactical choice turns out to be highly path dependent, such as the settings of peasant and rural rebellions where violence has high baseline capacity and scales meaningfully with turnout.

The parameters which separate these regimes map to observable features of the political, physical, and organizational environment – terrain, access to weapons, population density, and coordination infrastructure. The classification therefore generates predictions about which kinds of environments produce which kinds of tactical trajectories, contributing to longstanding literature on how the form of contention shifts with changes in social structure, urban

form, and communications technology (Beissinger 2022; Tarrow 1998; Tilly 1978), and offering a structural account of the widely documented success premium of nonviolent campaigns (Chenoweth and Stephan 2011).

Purely peaceful trajectories are only possible when the baseline power of violence is sufficiently low; once it is high, violence must appear somewhere on the equilibrium path. The success premium of nonviolence may therefore reflect not only a causal advantage of peaceful tactics, but also a selection effect: movements that *can* remain exclusively peaceful tend to operate in environments where success is more feasible to begin with. By contrast, there is no parameter region in which violence is unconditionally optimal. Because peace has a higher ceiling than violence, sufficiently pessimistic beliefs always make peace preferable in continuation. Hence, persistent violence is never a "structural" feature of the environment, but rather a contingent phenomenon.

This paper also offers a disciplined way to model repeated protests using global games (Morris and Shin 1998, 2003). While a large literature models protests and collective action using global games (Boix and Svolik 2013; Casper and Tyson 2014; Edmond 2013; Little, Tucker and LaGatta 2015; Rundlett and Svolik 2016; Shadmehr and Bernhardt 2017; Tyson and Smith 2018), extending the framework to dynamic settings is technically challenging because the interaction of accumulated private signals and the public signal of regime survival reintroduces multiplicity (Angeletos, Hellwig and Pavan 2007; Little 2017; Shadmehr 2019). Recent approaches address this through forward-looking altruism (Gieczewski and Koçak 2024) or staggered entry (Dasgupta 2007). This paper offers an alternative: agents receive private signals of cost rather than of regime strength, so beliefs about the regime remain common and are updated publicly through observed failures, whilst participation decisions are cutoff in cost. The result is preservation of uniqueness, as well as an attractively close relationship between each-period's cost threshold and contemporaneous public beliefs. The only other paper to the author's knowledge with a similar structure is Malashkhia and Malashkhia (2025), although their setup uses lognormal costs and does not use the global games selection argument.

More broadly, this papers offers a new theory of tactical choice in protest movements grounded in the interaction between learning, belief management, and the technology of contention. Existing models of contentious politics with tactical choice typically treat organizational capacity as exogenous or have tactical decisions made after participation decisions (Bueno de Mesquita 2013; Qiu 2022). By contrast, this model features a tight relationship between tactical choice, turnout, and beliefs: tactics shape the distribution of protest power, which determines what is learned from failure, which shapes future turnout and tactical choice.

The draft proceeds as follows. Section 2 briefly discusses related literature. Section 3 presents the model. Section 4 analyzes the model, abstracting away from tactical choice to establish the core mechanisms of cutoff participation, monotone belief decline, and the consequences of failure. Section 5 analyzes tactical choice. Section 6 concludes by discussing extensions to alternative leadership structures and other directions.

2 Literature

This paper speaks to three strands of existing work. The first is formal models of contentious politics with tactical choice. Morris and Shadmehr (2024) study repertoire choice in a one-shot setting, focusing on the decision to offer a wider or narrower set of actions, rather than the choice between tactics that differ in turnout sensitivity. Bueno de Mesquita (2013) models a similar tactical choice in a repeated setting, but organizational capacity is exogenously determined and per-period tactical decisions are made *after* participation decisions, so the feedback from tactic to turnout is absent. Qiu (2022) studies the choice between guerrilla and conventional strategies, but embeds it in a bargaining process between rebels and government rather than in a mobilization problem. Leventoğlu and Metternich (2018) treat protest size as a costly signal of rebel strength, but the signaling technology does not vary by repertoire and organizational strength is exogenous. None of these models feature the mechanism studied here where tactical choice is made *before* participation, and organizational capacity is endogenous to beliefs, so that the chosen tactic shapes turnout, turnout shapes what is learned from failure, and learning reshapes the environment that tactics operate in.

Second, the paper speaks to a large literature in comparative politics on the historical form and tactics of mass contention. Beissinger (2022) documents a historical shift from nineteenth-century urban revolutions to rural social revolutions in the early twentieth century, and then a re-urbanization of revolutionary episodes from the mid-twentieth century onward. Compared to the drawn-out rural campaigns they supplanted, modern urban revolutions unfold quickly and exhibit substantial apparent unpredictability – features that align with the model's core mechanisms of path-dependent belief updating amplified under turnout-sensitive tactics and muted under turnout-insensitive ones. More broadly, the repertoires-of-contention tradition (Tarrow 1998; Tilly 1978) emphasizes that the forms of collective action available to movements are shaped by social structure, communications technology, and the built environment. This paper formalizes that insight: by parameterizing how tactics convert turnout into power, the model connects observable environmental features to predicted patterns of tactical persistence and change. Much of the existing work implicitly treats "one protest, one tactic," by contrast, this paper is focused on tactical evolution *within* movements and explains when and why organizers switch.

Third, the paper draws somewhat on the logic of experimentation, albeit in a setting with a novel feedback structure. In canonical bandit problems, an agent chooses between a known "safe" arm and an unknown "risky" one, and the primary concern is whether agents experiment optimally – how quickly they learn the value of the risky arm relative to the safe one. The protest setting introduces a complication absent from that framework: the value of each "arm," or tactic, changes endogenously over time, because the effectiveness of each tactic itself depends on mobilization, which itself depends on beliefs shaped by prior outcomes. Hence, "experimentation" in this setting is not only about learning which tactic is better suited to the regime being contested, but about managing the beliefs that determine how much power each tactic can generate tomorrow. In effect, the organizer is choosing a "technology" while also endogenously changing how both technologies will operate in the future.

3 Model

Overview. There are two periods, $t \in \{1, 2\}$. In each period there may be a protest against a regime of unknown strength θ. Before the game begins, regime strength θ is drawn $Unif[m, M]$ with $M > 1$, constituting the common prior. Let F_t denote the commonly held beliefs about θ at the beginning of period t, so $F_1 = Unif[m, M]$.

Two tactics are available: *peaceful* and *violent*, denoted by subscripts p and v respectively.[1] Each tactic $\tau \in \{p, v\}$ is associated with a production function $A_\tau(\ell)$ that maps turnout $\ell \in [0, 1]$ (the share of eligible individuals who participate) into protest power. A protest succeeds if and only if $A_\tau(\ell) > \theta$. If the period-t protest succeeds, the game ends immediately; otherwise the game proceeds to the next period (and ends after period 2 in any case).

Tactics and success. For all values of turnout $\ell \in [0, 1]$, assume:

(A1) $A_v(0) > A_p(0)$: violence has higher baseline power,

(A2) $A_v(1) < A_p(1)$: peace has higher full-participation power,

(A3) $A'_v(\ell) < A'_p(\ell)$: violence is less turnout-sensitive at every level.

A1 captures the idea that violent or confrontational tactics can generate meaningful disruption even at low participation. A2 says that at full mobilization, peaceful tactics can generate greater total power, in other words, the "ceiling" of peace exceeds that of violence. A3 says

[1] The restriction to two tactics rules out simultaneous deployment of different repertoires by different factions. The stylization captures environments in which movements coordinate around a dominant tactic in each period, consistent with the historical observation that most movements exhibit a single predominant repertoire at any given time (Bueno de Mesquita 2013; Qiu 2022).

that each additional participant contributes more to peaceful power than to violent power. Collectively, these assumptions ensure that the two tactics cross exactly once: there exists a unique $\ell^* \in (0,1)$ such that $A_v(\ell) > A_p(\ell)$ for $\ell < \ell^*$ and $A_p(\ell) > A_v(\ell)$ for $\ell > \ell^*$.

Participation payoffs. Each agent i who participates in a period-t protest incurs the cost

$$c_{it} = c_t + \epsilon_{it},$$

where $c_t \sim \text{Unif}[\underline{c}, \bar{c}]$ is a common shock affecting all agents in period t, and $\epsilon_{it} \sim \text{Unif}[-\epsilon, \epsilon]$ is idiosyncratic. All shocks are drawn independently across agents and periods.[2] Assume $\underline{c} < 0$ and $\bar{c} > 0$, which guarantees two-sided limit dominance: for any beliefs, some agents always participate (those with sufficiently low costs) and some never do (those with sufficiently high costs). An agent who participates in a successful protest receives a private benefit normalized to 1.[3] An agent who does not participate, or who participates in a failed protest, receives zero.

Timing. Within each period:

1. The protest's tactic $\tau \in \{p, v\}$ is chosen.
2. Costs are realized; individuals decide whether to participate.
3. The protest occurs; turnout and the outcome are publicly observed.
4. If the protest succeeds, the game ends. If it fails and $t = 1$, proceed to period 2. If it fails and $t = 2$, the game ends.

The solution concept is Perfect Bayesian Equilibrium. In the baseline analysis, tactics are chosen by a success-maximizing organizer whose payoff is 1 if the protest succeeds at any point and 0 otherwise.

4 Analysis without tactical choice

Why do protest movements decline over time? Before studying how tactical choice can manage this decline, I first show that the model's core mechanics – the link between beliefs, participation, and learning – already produce a substantively rich theory of movement deterioration. Even with a single tactic, failed protests generate self-reinforcing pessimism: each unsuccessful

[2] The independence assumption abstracts from "core activists" with persistently low costs and from autocorrelated conditions such as sustained crackdowns. It ensures that expected turnout in period 2 depends only on updated beliefs and the participation cutoff, not on the period-1 cost realization, isolating the mechanism this paper studies: belief-driven path dependence through learning about regime strength.

[3] Agent payoffs also include a public benefit $\nu \cdot \mathbf{1}\{\text{success}\}$ received regardless of participation. This component plays no role in the baseline analysis, since agents condition participation solely on private costs and success probabilities, but will matter for extensions with welfare-maximizing organizers.

attempt reveals information about the regime's resilience, which depresses public confidence, which tightens participation, which weakens the expected power of the next attempt. The results in this section characterize this downward spiral formally and identify its key properties: the monotone erosion of public beliefs about regime strength, the inherent path dependence of movement trajectories, and the asymmetric damage inflicted by failures of different sizes. These properties operate regardless of which tactic is used; when tactical choice is reintroduced in the next section, its role is precisely to shape and mitigate these forces.

Fix a single production function $A(\ell)$ mapping turnout $\ell \in [0,1]$ to protest power. In each period t, a protest succeeds iff $A(\ell_t) > \theta$; the game ends immediately upon success. Otherwise, agents observe that the regime survived an attack of power $A(\ell_t)$, and the common posterior F_{t+1} is obtained by truncating F_t from below:

$$F_{t+1}(x) = \begin{cases} 0, & x < A(\ell_t), \\ \dfrac{F_t(x) - F_t(A(\ell_t))}{1 - F_t(A(\ell_t))}, & x \geq A(\ell_t) \end{cases} \tag{1}$$

The posterior concentrates on regime types that could have survived the observed protest, shifting probability mass toward stronger regimes.

Cutoff participation and uniqueness

The first result establishes how public beliefs translate into individual participation decisions.

Lemma 1 (Cutoff participation)**.** *There exists a threshold $\hat{c}_t$ such that agent i participates in the period-t protest if and only if $c_{it} < \hat{c}_t$, where*

$$\hat{c}_t = \int_0^1 F_t(A(\ell))\, d\ell \tag{2}$$

The derivation proceeds in three steps. First, an agent with realized cost c_{it} participates whenever the probability of success exceeds the cost, i.e. $\Pr(\text{success} \mid F_t) > c_{it}$. The marginal agent, with cost exactly $\hat{c}_t$, is indifferent: $\Pr(\text{success} \mid F_t) = \hat{c}_t$.

Second, the probability of success depends on turnout, which the marginal agent does not observe directly. Given turnout ℓ_t, the probability that the regime falls is $F_t(A(\ell_t))$. Since the marginal agent does not know the common cost shock c_t, she must average over the turnout that different values of c_t would induce:

$$\hat{c}_t = \int F_t(A(\ell(c_t)))\, g(c_t \mid c_i = \hat{c}_t)\, dc_t.$$

I now apply a standard observation from the global games literature. The marginal agent's only uncertainty is where she sits in the cost distribution relative to the population. As the

common shock c_t shifts, the fraction of agents below the cutoff slides smoothly from 0 to 1. From the marginal agent's perspective, all such fractions are equally likely: the change of variables $c_t \mapsto \ell_t$ yields a uniform density on $[0, 1]$.[4] The indifference condition therefore simplifies to

$$\hat{c}_t = \int_0^1 F_t(A(\ell))\, d\ell,$$

which is equation (2). The participation threshold equals the average probability of success, integrated over all possible turnout levels.

The cutoff structure immediately delivers uniqueness:

Proposition 1 (Unique cutoff PBE)**.** *Fix any realization of the common shocks $\{c_t\}_{t=1}^2$. There exists a unique Perfect Bayesian Equilibrium in threshold strategies. In each period t, agents participate if $c_{it} < \hat{c}_t$ with $\hat{c}_t$ given by* (2)*; turnout is determined by the mass of agents below the cutoff; and beliefs update by truncation after failure. Turnout and success probabilities are unique conditional on $\{c_t\}$.*

The intuition is immediate. Lemma 1 pins down a unique $\hat{c}_t$ for any beliefs F_t, and hence a unique turnout ℓ_t for any realization of c_t. Belief updating is deterministic given the observed outcome. Backward induction over the two periods completes the argument.

Equation (2) makes transparent the link between beliefs and mobilization. More pessimistic beliefs – F_t that place greater weight on strong regimes – reduce $F_t(A(\ell))$ pointwise, lowering the integral and tightening the cutoff. Since turnout consists of all agents with $c_{it} < \hat{c}_t$, lowering the cutoff reduces turnout for every realization of the common shock. Worse beliefs produce lower thresholds, which produce lower expected turnout, which (upon failure) produces worse beliefs (in a FOSD sense.) The following results formalizes the decline of beliefs and the accompanying (expected) decline of turnout:

Corollary 1 (Monotone decline of thresholds)**.** *The cost threshold is weakly decreasing over time: $\hat{c}_{t+1} \leq \hat{c}_t$.*

Corollary 2 (Turnout declines in expectation)**.** *For any distribution of common shocks with c_t independent of past outcomes, $\mathbb{E}[\ell_{t+1}] \leq \mathbb{E}[\ell_t]$, with strict inequality whenever the game continues after a nontrivial failure.*

Note that expected decline does not preclude realized increases. A fortunate draw of the common shock can produce a turnout surge even after a failed first period. But on average, the movement is weaker tomorrow than today – not because participants have lost interest, but because they have learned that the task is harder than they thought.

[4] This holds for any continuous, strictly increasing idiosyncratic noise distribution, not only the uniform.

A natural question is whether some levels of mobilization are worse for a movement's future than others. Conditional on failure, the answer is unambiguous: larger protests are more damaging to future beliefs.A sizable showing that falls short reveals more about the regime's resilience than a small one, driving a sharper pessimistic revision and depressing future willingness to participate – a failed march of fifty thousand is far more demoralizing than a failed gathering of fifty. However, this does not mean that a movement hoping to maximize its chances at success would prefer low turnout to intermediate turnout. A larger protest is also more likely to succeed outright, and this direct benefit turns out to compensate for the harsher consequences of failure. Formally, the *eventual* success function $H(x) := \Pr(\text{success by period 2} \mid \text{period-1 power} = x)$ is weakly increasing, indicating that higher realized power always improves total lifetime prospects, despite the fact that the continuation value conditional on failure is decreasing. The risk to beliefs is always offset by the higher probability of never needing a second attempt.

The tension between these dual objectives – the desire to achieve early success, which is more likely with larger protests, versus the desire to minimize damage done to future beliefs, which is also exacerbated by larger protests – is most keenly felt at low-to-moderate levels of mobilization. In this range, the probability of outright success is small, but the informational damage from failure is already substantial. At high mobilization, the tension is moot: the movement is likely to win, if victory is feasible at all. In other words, the eventual success function, $H(x)$, displays increasing returns, implying that the marginal gain to additional turnout is smallest when the movement is most fragile. In the single-tactic setting, movements have no tool with which to manage this tension: they must simply live with whatever mobilization arises and its attendant consequences. When tactical choice is reintroduced, however, organizers gain some leverage over the asymmetry between low- and high-mobilization consequences for eventual success. Violent or disruptive tactics, with high baseline power and low turnout sensitivity, *compress* the range of possible outcomes. Peaceful or mass tactics with no baseline power but high turnout sensitivity do the opposite, spreading probability over a wider range of realizations.

Hence, when organizers choose tactics, they are in effect choosing distributions over realized power *and over post-failure belief states*. Choosing peaceful tactics amounts to gambling on high realized power at the cost of more informative failures when mobilization falls short, while choosing violent tactics reduces both the upside from very high turnout and the downside from highly revealing failures. In the next section, I characterize precisely how organizers use this lever, and in particular how forward-looking concerns about failure shape tactical choice.

5 Tactical choice

Reintroducing tactical choice raises a simple but nontrivial question: in what order should a movement use tactics that differ in baseline power and turnout sensitivity? A natural first guess is that an organizer seeking to maximize success should begin with peaceful mass tactics, when public optimism and expected turnout are still relatively high, and turn to violence only later, once prior failures have depressed beliefs and participation. Alternatively, one might think that if violence has especially high baseline power, then leading with it should be attractive: optimistic beliefs still help mobilization, while violence's floor guarantees a relatively strong showing. But the problem is not so simple. As discussed before, the organizer is choosing not only how much expected power to generate today, but also which post-failure belief states the movement is most likely to face tomorrow.

To clarify incentives, I focus on a success-maximizing organizer who chooses the movement's tactic in each period to maximize the probability of achieving success at any point in the movement's lifetime. The organizer controls which tactic is used, but does not pre-commit to a plan: each period's choice is a best response to the history and information available at that point. Nor does the organizer directly control turnout. Protesters observe the chosen tactic and the realization of shocks, then decide individually whether to participate. In that limited sense, the organizer's problem resembles one of experimentation: choosing a tactic today simultaneously determines the likely power of tactics used in the future.

Two benchmark results help to fix ideas before turning to the richer intermediate cases. The first benchmark result identifies conditions under which peace is optimal throughout:

Proposition 2 (A sufficient condition for peace on the entire path)**.** *Suppose peace is optimal in the continuation problem, i.e.* $\hat{c}_2^p(x) \geq \hat{c}_2^v(x)$ *for all* x*. Define*

$$H(x) := F_1(x) + (1 - F_1(x))\hat{c}_2^p(x),$$

so that period-1 expected success under tactic $\tau \in \{p, v\}$ *is* $W_1^\tau = \int_0^1 H(A_\tau(\ell))\, d\ell$*. If, for every threshold* x*,*

$$\int_0^1 (A_p(\ell) - x)_+ \, d\ell \;\geq\; \int_0^1 (A_v(\ell) - x)_+ \, d\ell, \tag{3}$$

then $W_1^p \geq W_1^v$*, so peace is optimal in both periods.*

The result requires not only that peace is attractive in the continuation, but also that peace is sufficiently advantaged that there is no *anticipatory* reason to use violence in the first period to engineer more favorable continuation states. Even if peace is preferred to violence in the continuation problem for *every* posterior belief, it is not necessarily true that peace should be used first. Violence could be worth using first if it is especially good at steering the movement

away from low-power failures and toward continuation states in which peace remains effective.

Consider the following example: Suppose peace is always the preferred period-2 tactic, but continuation value rises sharply with realized first-period power. If peace in period 1 generates a very dispersed set of outcomes – occasionally very high power, but also a substantially larger risk of very weak failures – while violence generates a more reliable intermediate level of power, then violence may be optimal in period 1 precisely because it avoids disastrous failures and thereby delivers more favorable continuation states for period-2 peace. Thus, peace being optimal in continuation is not, by itself, sufficient to imply peace on the entire path.

Finally, while there are parameter regions in which peace is preferable at every reachable posterior belief, the same is not true for violence. Put simply, if the peaceful tactic can reach higher maximum power than violence, then there are always some beliefs for which peace strictly dominates violence in the continuation. To be clear, this does *not* imply that violence will never be used on the equilibrium path, only that repeated use of violence is path dependent.

The second benchmark identifies the opposite extreme: violence with arbitrarily high baseline power but no turnout sensitivity at all. Such a tactic may still be useful as a fallback, but it is never optimal to deploy in the first period. This clarifies exactly when organizers are willing to sacrifice an (arbitrary) amount of myopic power in favor of beliefs that favor turnout-sensitive tactics. A "flat" tactic generates the same informative failure regardless of turnout, depressing beliefs without the compensating upside of occasionally converting high mobilization into success. Since such a tactic can always be held in reserve for period 2 with no expected loss in power – when belief management no longer matters – there is no reason to sacrifice period-1 option value by deploying it early.

Lemma 2. *If $A_v(\ell) \equiv \bar{v}$ for all ℓ, then a success–maximizing organizer never uses violence in period 1.*

This "flat violence" case demonstrates the most extreme version of a case in which an organizer prefers to sacrifice an *arbitrary* amount of myopic win probability in the first period. However, a natural question is the robustness of this strong result: is it knife-edge or more general? Indeed, while these benchmark results characterize extremes generated by particular forms of the tactic functions, in reality, most repertoires of contention lie in between. To characterize these intermediate environments and their accompanying implications for patterns of tactical choice, I now impose a linear parameterization that makes the tradeoff between baseline violent power and violent turnout sensitivity transparent.

Linear parameterization

I now study the case in which tactics are linear. In particular, I assume $A_p(\ell) = \ell$, $A_v(\ell) = \alpha + \beta\ell$, with $\alpha + \beta < 1$ to satisfy the assumption that $A_p(1) > A_v(1)$. The parameterization is without loss of generality within the class of linear production functions.[5]

The main result shows that tactical choice is cutoff in both periods. In the second period, the organizer prefers violence if and only if the realized power of first-period protest falls below a threshold $c(\alpha, \beta)$: violence is the preferred continuation tactic after sufficiently weak first-period protests. In the first period, the organizer prefers violence if and only if β – the turnout responsiveness of the violent tactic – exceeds a threshold $\beta^*(\alpha)$.

Proposition 3 (Tactical choice with linear functions.)**.** *The following results dictate tactical choice in both periods:*

(i) *(*Period-1 cutoff.*) At $\beta = 0$, $W_1^p(\alpha, 0) > W_1^v(\alpha, 0)$ (Lemma 2), while at $\beta = 1 - \alpha$, $W_1^v(\alpha, 1-\alpha) > W_1^p(\alpha, 1-\alpha)$. Moreover, $\Delta(\alpha, \beta) := W_1^v(\alpha, \beta) - W_1^p(\alpha, \beta)$ is continuous and strictly increasing in β on $(0, 1-\alpha)$. Consequently, there exists a unique $\beta^*(\alpha) \in ((1-\alpha)^2, 1-\alpha)$ such that*

$$W_1^v(\alpha, \beta) > W_1^p(\alpha, \beta) \iff \beta > \beta^*(\alpha).$$

(ii) *(*Period-2 cutoff.*) After a period-1 failure at realized power $x \in [0, 1]$, the organizer chooses violence in period 2 if and only if $x \leq c(\alpha, \beta)$, where*

$$c(\alpha, \beta) = \begin{cases} 0, & \beta \leq 1 - 2\alpha, \\ \sqrt{2\alpha + \beta - 1}, & 1 - 2\alpha < \beta \leq (1-\alpha)^2, \\ \dfrac{\alpha + \beta - \sqrt{\beta}}{1 - \sqrt{\beta}}, & \beta > (1-\alpha)^2. \end{cases}$$

In addition, $c(\alpha, \beta)$ is weakly increasing in both α and β on the feasible set.

The interaction of the two cutoffs partitions the feasible parameter space into three qualitatively distinct tactical regimes, ordered below by the turnout responsiveness of violence (lowest to highest) – an ordering that also tracks increasing path dependence.

1. **Exclusive peace.** $\beta < \beta^*(\alpha)$ and $\beta \leq 1 - 2\alpha$ (so $c(\alpha, \beta) = 0$).
   Peace is used in period 1 and remains optimal in period 2 for all realizations. Violence is never attractive, either as an opening tactic or as a fallback.

[5] Take any pair $A_p(\ell) = a_p + b_p\ell$ and $A_v(\ell) = a_v + b_v\ell$ satisfying the model's assumptions ($a_v > a_p$, $a_v + b_v < a_p + b_p$, $b_v < b_p$). A straightforward change of variables $\tilde{\theta} = (\theta - a_p)/b_p$ maps regime strength into a rescaled space in which $\tilde{A}_p(\ell) = \ell$ and $\tilde{A}_v(\ell) = \alpha + \beta\ell$, where $\alpha = (a_v - a_p)/b_p > 0$ and $\beta = b_v/b_p \in (0, 1)$.

2. **Peace-first with learning-dependent escalation.** $\beta < \beta^*(\alpha)$ and $\beta > 1 - 2\alpha$ (so $c(\alpha, \beta) > 0$).
   Peace is used first. After a period-1 failure, the organizer switches to violence if and only if the failure is sufficiently weak, i.e. if realized power falls below $c(\alpha, \beta)$. Such low-power outcomes are uninformative about regime strength, making violence's guaranteed floor attractive relative to the risk of another feeble peaceful showing.

3. **Violence-first with learning-dependent continuation.**[6] $\beta > \beta^*(\alpha)$ and $\beta > (1-\alpha)^2$ (equivalently $\alpha < c(\alpha, \beta) < \alpha + \beta$).
   Violence is used first. After a period-1 failure, the organizer remains violent if the failure is sufficiently weak, and switches to peace otherwise. Both persistence and switching occur on the equilibrium path, depending on what the first period reveals.

A fourth logical possibility – violence-first with predetermined switching to peace – is empty under success maximization, since the parameter values at which it becomes optimal to lead with violence are exactly those for which continuation choice is already history-dependent.[7]

Regime 1 describes environments in which peaceful and violent tactics are both fundamentally mass technologies. Violence has no meaningful baseline capacity: there is no level of coercive or disruptive power the movement can generate without broad participation. This rules out violence as either a first-period tactic (its expected power is too low to justify forgoing peace's upside) or a continuation fallback (its floor offers no insurance against weak showings). Substantively, these are environments in which even disruptive or violent action remains highly dependent on turnout – movements operating in open urban terrain with wide boulevards and redundant infrastructure rather than ones able to exploit rugged geography, key chokepoints, or limited arterial roads; movements facing strong moral constraints on injurious force; or movements contesting issues for which coercion is simply not an effective lever. The common thread is that both tactics live or die by how many people show up.

Mathematically, Regime 1 is the linear analogue of Proposition 2. It is straightforward to verify that the sufficient conditions are satisfied. As α rises, however, the conditions become harder to satisfy, and the regime disappears entirely once $\alpha > 1/2$. At that point, every admissible parameter configuration involves the possibility of violence appearing somewhere on

[6] This regime encompasses the possibility of *near*-exclusive violence. As $\alpha + \beta \nearrow 1$, the fraction of violence-first histories that sustain violence into period 2, given by $(c(\alpha, \beta) - \alpha)/\beta$, approaches 1, so switching to peace becomes increasingly rare, but never vanishes entirely. At $x = \alpha + \beta$, peace strictly dominates violence by $(1 - \alpha - \beta)^2/2 > 0$, so $c(\alpha, \beta)$ remains strictly below $\alpha + \beta$. Near-exclusive violence is therefore a limiting case of this regime as the ceiling of the violent tactic approaches that of peace.

[7] Formally, this requires showing that $\beta^*(\alpha) > (1 - \alpha)^2$ for all $\alpha \in (0, 1)$, which is established in the proof of Proposition 3.

the equilibrium path. Under success-maximizing leadership, purely peaceful movements cannot arise in environments where violence carries substantial baseline capacity. Purely peaceful trajectories require that violence be a fundamentally crowd-dependent technology.

The "color revolutions" – Georgia (2003), Ukraine (2004), Kyrgyzstan (2005), and the Euromaidan movement in Ukraine (2013–14) illustrate this logic as a class. In each case, the movement's leverage derived almost entirely from the visible willingness of large numbers of ordinary citizens to occupy central urban spaces: open squares, broad avenues, government precincts. There was no insurgent tradition, no meaningful access to weapons, and no terrain that would have allowed a small vanguard to exert disproportionate force. The Euromaidan case is particularly instructive: even when violent clashes did emerge late in the crisis, they were themselves mass events, rather than small-numbers insurgency. Both tactics remained crowd-dependent, exactly as Regime 1 requires.

The logic connecting physical environment to tactical viability has not been lost on state actors. The Haussmannian renovation of Paris in the 1850s and 1860s is perhaps the most deliberate historical example of an intervention on α. The medieval street network that had enabled decades of effective barricade insurgency in narrow strets, dense constructions, and numerous chokepoints, was dismantled and replaced with wide boulevards, open sightlines, and redundant arterial roads. The renovation destroyed precisely the terrain features that had given small numbers of insurgents disproportionate leverage, converting Paris from an environment in which violence carried substantial baseline power into one in which both tactics were fundamentally crowd-dependent. In the language of the model, Haussmann engineered a downward shift in α large enough to move Paris into Regime 1 – and the subsequent history of Parisian contention, dominated by mass demonstrations rather than barricade warfare, is consistent with that prediction.

Regime 2 describes what is arguably the most empirically recognizable pattern: peace-to-violence escalation. The organizer begins with peaceful tactics, but after sufficiently disappointing first-period outcomes, pivots to violence. This arises when $\beta > 1 - 2\alpha$ (so that $c(\alpha, \beta) > 0$) but $\beta < \beta^*(\alpha)$ (so that peace is still preferred in period 1). The organizer leads with peace because it offers higher upside when the movement's credibility and expected turnout are still intact. Violence becomes appealing only after a weak showing – one that fails to topple the regime but is uninformative enough to leave the organizer believing the regime might still be relatively weak. In that circumstance, violence's floor provides a guaranteed level of power that avoids a repeat of the feeble showing that a poorly attended peaceful protest produces.

The regime is predicated on violence possessing an intermediate level of turnout respon-

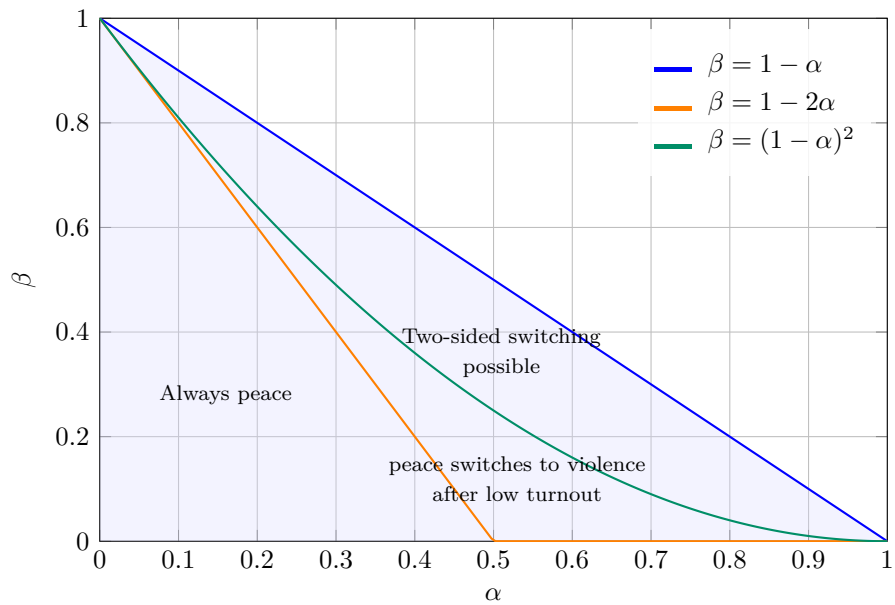

Figure 1. Cutoffs and tactical choice. The feasible parameter region is shaded in light blue. This figure summarizes the continuation regimes; the period 1 boundary $\beta^*(\alpha)$ lies somewhere strictly above the green line $\beta = (1 - \alpha)^2$ and is omitted for clarity.

Left of the orange line, peace is used in both periods. Between the green and orange line lies the parameter region in which peace is the preferred starting tactic, and the violent tactic can be used following a low peaceful turnout. Between the green and blue line, path dependence following both peace and violent starts are possible.

siveness, and maps onto a wide array of cases. Many movements begin with marches, strikes, and demonstrations, and escalate to property destruction, riots, or armed resistance only after early peaceful efforts yield weak results. Environments that produce this pattern tend to feature limited baseline violent capacity, but meaningful crowd-based violent options: settings where the movement has access to rioting, barricade construction, or improvised resistance, but no standing insurgent infrastructure.

The main prediction in this regime is that escalation is driven by ambiguity, not by desperation or confirmed weakness. A naive account of path-dependent escalation which posits that "deterioration leads to violence" misses the point about *ambiguity*. After low-power failures, protesters believe that the regime might still be beatable, but peace clearly is not the tool to beat it with at this level of mobilization. A massive peaceful march that fails to dislodge the regime is discouraging, but the movement is likely to stick with peace; small, inconclusive turnouts push the organizer toward violence's floor. Escalation to violence is a sign of *unresolved uncertainty*, rather than of confirmed regime weakness.

The Egyptian revolution of 2011 exhibits within-case variation consistent with model predictions. In Cairo, where turnout on January 25 was overwhelming, protests remained largely peaceful through the decisive days – the sheer scale of mobilization made violence unnecessary and peace maximally powerful. In smaller cities where initial demonstrations drew disappointing crowds, protesters escalated to violent clashes with police, storming and burning government buildings. The escalation was not uniform across the movement; it tracked the local informativeness of first-period turnout.

Regime 3 arises when violence is sufficiently responsive to turnout that it is both an appealing opening tactic *and* a potentially appealing continuation option. The organizer leads with violence, but failures are contingent. As with the previous case, low-turnout violent failures remain ambiguous: a weak showing can be attributed to insufficient mobilization rather than regime resilience, leaving the organizer willing to try violence again. Similarly, when a powerful protest produced by the violent tactic results in failure, the movement updates decisively toward a strong regime and pivots to peace, the higher-ceiling tactic.

These cases may correspond to sustained insurgencies or armed campaigns in which the movement persists with violence through initial setbacks, interpreting early failures as reflecting inadequate coordination or resources rather than fundamental futility. The movement abandons violence only after a major, well-supported operation is decisively defeated. The environments producing this regime tend to feature both substantial baseline violent capacity *and* high scalability: dense urban populations near key targets (where mass participation in violent action overwhelms security forces), strong coordination infrastructure, and violent tactics that are inherently crowd-dependent, such as riots, mass seizures of territory, or coordinated urban uprisings rather than targeted sabotage.

The Syrian uprising's trajectory may reflect this logic of this third regime. What began as mixed protests in 2011 escalated into sustained armed insurgency, with rebel factions persisting through repeated setbacks. Early losses were interpreted as reflecting insufficient coordination, inadequate weapons supply, or unfavorable local conditions rather than the fundamental resilience of the Assad regime. The armed strategy was abandoned only after major offensives (notably in Aleppo) were decisively defeated by a regime backed by foreign military intervention, at which point the lesson about regime strength became unavoidable. Of course, external intervention and factional fragmentation complicate direct mapping to the stylized model here, but the broad pattern of persistent violence through ambiguous failures, followed by de-escalation after decisively large defeats fits the model logic.

The comparison across regimes highlights the central role of violence's *floor* in determining where path dependence can arise. A nontrivial cutoff $c(\alpha, \beta) > 0$ emerges whenever violence is responsive enough to turnout that different first-period realizations *could* lead to different continuation tactics. But a positive cutoff is necessary, not sufficient, for path dependence. The reason is that different first-period tactics produce different ranges of realized power. After a peace-first start, power is in the range $[0, 1]$, so any positive cutoff creates histories on both sides of it: some that trigger escalation and some that do not (Regime 2).

After a violence-first start, however, realized power is at least α. Path dependence following a violent start therefore requires the cutoff to exceed the floor: $c(\alpha, \beta) > \alpha$. This is the condition that separates Regime 3, where both persistence and switching are possible, from the region where a violence-first path would always transition to peace. In this sense, Regime 3 is the region of maximal variation in tactical trajectories – the parameter space in which the movement's path is most sensitive to what the first period reveals.

Comparative statics

Having characterized these regimes, I now turn to whether expand or shrink as α and β vary. Increasing α or β corresponds to moving in a northeasterly direction across the feasible range of Figure 1. As α rises, exclusive peace becomes harder to sustain and eventually impossible ($\alpha > 1/2$); as β rises, path dependence becomes more common and violence becomes viable as an opening tactic. To lend substantive weight to these observations, I briefly discuss some the factors that may determine α and β.

α, the baseline power of violence – the coercive or disruptive capacity a movement can generate without mass participation – depends on the physical and organizational environment. Several features push α upward:

- *Terrain and infrastructure.* Mountainous or densely forested terrain, narrow arterial roads, and key infrastructural chokepoints (bridges, tunnels, power stations) allow small numbers to exert disproportionate leverage. A handful of insurgents can shut down a supply route or hold a pass that would require thousands of peaceful marchers to blockade symbolically.

- *Access to weapons and materiel.* Movements with access to arms, explosives, or the technical capacity for sabotage have higher baseline violent power. This access may come from state sponsors, captured stockpiles, cross-border flows, or the movement's own organizational history. Movements without such access – student movements, labor campaigns, pro-democracy movements in tightly policed states – have low α almost by default.

- *Organizational depth.* Experienced insurgent cadres, clandestine cell structures, and specialized military training raise α by making small-numbers violence more effective. A movement with a trained armed wing can mount meaningful operations with minimal turnout; a movement that depends on spontaneous crowd action cannot.

β, the turnout sensitivity of violence – how much additional participation amplifies violent power – depends on different features:

- *Urban density and proximity to targets.* When large populations live near administrative centers, military installations, or symbolic targets, mass participation in violent action translates directly into overwhelmed security forces and seized territory.
- *Communication and technology.* Networks that facilitate rapid mobilization – social media, encrypted messaging, preexisting organizational structures – raise β by making it easier to convert willing participants into coordinated violent action. When coordination is difficult, additional willing participants may not translate into additional effective power.
- *Nature of the violent tactic.* Some forms of violence are inherently more scalable than others. Riots and mass seizures of territory scale with turnout; targeted assassinations and infrastructure sabotage likely do not. A movement whose "violent tactic" is rioting has high β; one whose violent tactic is a car bomb has low β and high α.

To provide mathematical intuition behind the broad observation highlighted above, I delve into the comparative static on each object separately below:

The period-2 cutoff $c(\alpha, \beta)$ is weakly increasing in both arguments. Raising α expands the set of belief states under which violence is optimal as a fallback: higher baseline capacity makes it worthwhile to rely on violence after a wider range of discouraging outcomes. Raising β strengthens crowd complementarities for violence, making it more attractive to persist with violence following a weak showing rather than pivoting to peace. In environmental terms, both channels push in the same direction: movements with access to more powerful or more scalable violent options should exhibit more frequent tactical switching (after peaceful starts) and more persistent violence (after violent starts).

The period-1 threshold $\beta^*(\alpha)$ is strictly positive for all $\alpha \in (0, 1)$ and strictly decreasing in α. The positivity result is a direct consequence of Lemma 2: no matter how powerful violence is at baseline, some turnout sensitivity is always required before leading with it, because a completely flat tactic sacrifices the option value of learning. The monotonicity operates through two reinforcing channels. The direct channel is that higher α raises violence's expected period-1

power. The indirect channel is that higher α expands the set of continuations in which violence remains optimal in period 2 (since $c(\alpha, \beta)$ is increasing in α), so the organizer anticipates that a violence-first start is more likely to be strategically sustainable – that is, violence may remain viable even after a disappointing opening. In applied terms, movements operating in environments with substantial baseline violent capacity need less scalability of violence before leading with it.

Moreover, the eventual-win-probability advantage of leading with violence is strictly increasing in β across the entire admissible range. There is no "sweet spot" for the scalability of violence beyond which further turnout responsiveness becomes counterproductive. The reason is that higher β helps on both margins: it raises violence's direct period-1 power *and* expands the set of continuations where violence remains useful. This means the period-1 threshold $\beta^*(\alpha)$ is crossed exactly once: once violence becomes the preferred opening tactic, making it even more scalable only strengthens the case for leading with it.

I close with a brief comment about the parameterization. As we have seen, the linear specification delivers closed-form regime boundaries, monotone comparative statics, and a conceptual mapping from (α, β) to observable features of the environment. If we were to study the more general class of nonlinear production functions, the eventual success function H would remain well-defined and weakly increasing, guaranteeing existence of a threshold separating violence-first and peace-first parameter regions. However, uniqueness of the threshold, closed-form characterization regime boundaries, and monotone comparative statics would likely be impossible to obtain.

Observable implications

The mapping from environments to regimes facilitates several testable predictions, some of which distinguish it from existing accounts of tactical choice in contentious politics.

The sharpest prediction concerns the *direction* of tactical switching. The model predicts that peace-to-violence escalation should follow weak, low-turnout first-period outcomes, while violence-to-peace de-escalation should follow strong, high-turnout outcomes that nonetheless fail. By contrast, existing accounts of escalation emphasize repression, frustration, or the failure of moderate tactics in general terms. The model emphasizes that this may only be true for certain *kinds* of failure. A massive peaceful protest that fails to dislodge the regime is discouraging but does not trigger a switch to violence; a poorly attended one does. Conversely, a small-scale violent action that fizzles does not push the movement toward peace; a large, well-supported violent action that is decisively defeated does. The variable that governs switching is not whether the movement failed, but *how much power the movement generated when it failed.*

This prediction is directional and, in principle, testable with protest-level data. The empirical strategy would require measurement of tactical choice at the event level, protest size or intensity for each event, and the ability to observe sequential tactical decisions within the same campaign. The prediction is that, conditional on failure, larger peaceful protests should be *less* likely to be followed by escalation to violence, while larger violent actions should be *more* likely to be followed by de-escalation to peace. The key empirical challenge, besides securing event-level data in sequence, is distinguishing the model's prediction (switching is driven by the *informativeness* of failure) from alternatives in which switching is driven by the *severity* of repression following failure.

Second, the incidence of path-dependent tactical switching should correlate with the turnout scalability of violence. Environments where violence is available but does not scale with participation (high α, low β) should exhibit rigid tactical sequences. More path dependence requires that violence be turnout-sensitive. Empirically, this requires that data distinguish between the *kind* of voilent tactic(s) primarily available to movements: are these options that scale with mass participation (riots, seizures of territory) or ones that at a fixed capacity (sabotage, assassination)? Datasets that distinguish these features should find that tactical switching is concentrated among movements with scalable violent options.

Third, the model predicts that changes in the informational environment should have asymmetric effects across regimes. Increased observability of protest outcomes – live-streamed confrontations, ubiquitous video, real-time social media reporting – exacerbate the effects that failures have on posterior beliefs about regime strength by making turnout and consequences more visible to all potential participants. The model predicts that this amplification is most consequential in Regimes 2 and 3, where tactical choice is responsive to information: greater observability should widen the gap between the tactical responses to weak versus strong failures. In Regime 1, where peace dominates regardless of what failures reveal, the informational environment is irrelevant to tactical choice. This suggests that the spread of information technology should produce more *varied* tactical trajectories within the regimes where switching is possible, without affecting the boundaries of the exclusively peaceful regime.

6 Discussion

The analysis throughout takes the perspective of a unitary organizer who chooses tactics solely to maximize the probability of toppling the regime. This is a strong assumption, but it is also what makes the benchmark useful. It isolates the strategic logic of tactical choice under uncertainty: how organizers trade off immediate effectiveness against the informational conse-

quences of failure, holding fixed the internal politics of the movement and the distributional costs of contention.

The most natural extension is to relax the organizer's objective. A *welfare*-maximizing leader would care not only about whether the regime falls, but also about the costs borne by participants through arrests, injuries, repression, and prolonged mobilization. Introducing those concerns would generally break the tight equivalence between success maximization and optimal tactical choice developed here. In particular, the cumulative costs of drawn-out failed contention may make organizers more willing to favor tactics that maximize short-run success, similar to introducing impatience through discounting. The benchmark result that flat violence is never used first will certainly no longer hold.

A second class of extensions changes not the organizer's objective, but who effectively chooses tactics. In particular, if participants are tactic-specific (e.g. peaceful protesters are drawn from a different population than violent protesters) the model should exhibit a tension between protesters wanting to hold onto the current tactic to "incur private benefits" versus switch tactics in order to offload costs to a different population. Alternatively, if participants update beliefs differently than on-participants, the citizens who show up in period 2 are those whose beliefs are still optimistic enough to justify participation. If the movement's surviving participants are disproportionately drawn from its most committed or radical segment – because moderates dropped out after a discouraging first period – then citizen-controlled tactical choice may exhibit an endogenous radicalization dynamic, similar to Gieczewski and Kosterina (2024): the surviving population prefers violence not because it is strategically optimal, but because the kind of person who still shows up is the kind of person who prefers violence.

Finally, real movements often feature multiple leaders who disagree over tactics. In such settings, tactical choice is the outcome of bargaining rather than unilateral decision. The relevant questions – who has veto power, whether factions can act unilaterally, whether bargaining failure leads to inaction or fragmentation – are rich enough to constitute a separate investigation. The current model provides the success-maximizing benchmark against which distortions from internal bargaining can be measured.

Another potential direction for the paper considers the role of violent flanks that exist alongside the main movement. This direction would emphasize that movements often need to distribute their capacity across different wings or factions which prefer different tactics. This would shift the analysis from tactical choice by a unitary organizer to strategic interaction across wings of the movement. A violent flank can alter the peaceful wing's prospects even when the main organizer does not choose violence directly: it may raise the movement's baseline

coercive capacity, change what failures reveal about regime strength, or reshape turnout by deterring moderates while energizing more radical participants. In that sense, violent flanks may either insure the movement against weak peaceful showings or instead undermine the very mass participation on which peaceful tactics depend. Extending the model in this direction would therefore offer a way to formalize when radical flanks complement the main movement and when they instead distort its tactical path away from what a unified success-maximizing organizer would choose.

Conclusion

This paper develops a model of dynamic tactical choice in protest movements, showing that tactical trajectories are governed by the *type* of failure a movement experiences, not merely by whether it has failed. Low-power failures push movements toward violence's guaranteed floor; high-power failures push them toward peace's higher ceiling. The linear parameterization maps these forces to observable features of the environment, generating testable predictions about the incidence and direction of tactical switching. By assuming a unitary success-maximizing organizer, it isolates informational and strategic logic from the distributional and political considerations that also shape real movements. The extensions sketched above suggest that relaxing this assumption will introduce systematic biases, but the direction and magnitude of these biases are open questions.

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Proofs Appendix

Proof of Proposition 2

Proof. The period-2 claim is immediate from the assumption that $\hat{c}_2^p(x) \geq \hat{c}_2^v(x)\,\forall x$. Under this assumption, the period-1 problem becomes

$$W_1^\tau = \int_0^1 H(A_\tau(\ell))\,d\ell, \qquad H(x) = F_1(x) + (1 - F_1(x))\hat{c}_2^p(x).$$

Using the peace continuation rule,

$$(1 - F_1(x))\hat{c}_2^p(x) = \int_0^1 \big(F_1(A_p(\ell)) - F_1(x)\big)_+\,d\ell,$$

so

$$H(x) = F_1(x) + \int_0^1 \big(F_1(A_p(\ell)) - F_1(x)\big)_+\,d\ell.$$

Differentiating gives

$$H'(x) = f_1(x)G_p(x) \quad \text{where} \quad G_p(x) = \int_0^1 \mathbf{1}\{A_p(\ell) \leq x\}\,d\ell.$$

Since f_1 is constant (uniform prior) and G_p is weakly increasing, $H'(x)$ is weakly increasing, so H is weakly increasing and convex.

Now let $\bar{A} := \max\{\sup_\ell A_p(\ell), \sup_\ell A_v(\ell)\}$. Since H is convex,

$$H(y) = H(0) + H'(0)y + \int_0^{\bar{A}} (y - x)_+\,H''(x)\,dx$$

for every $y \in [0, \bar{A}]$. Applying this with $y = A_\tau(\ell)$ and integrating over ℓ yields

$$W_1^\tau = H(0) + H'(0)\int_0^1 A_\tau(\ell)\,d\ell + \int_0^{\bar{A}} \left[\int_0^1 (A_\tau(\ell) - x)_+\,d\ell\right] H''(x)\,dx.$$

Equation (5) implies that the bracketed term is weakly larger for peace than for violence for every x. In particular, setting $x = 0$ also gives

$$\int_0^1 A_p(\ell)\,d\ell \geq \int_0^1 A_v(\ell)\,d\ell.$$

Since $H'(0) \geq 0$ and $H''(x) \geq 0$, each term in the decomposition is weakly larger under peace. Therefore $W_1^p \geq W_1^v$, i.e. peace is optimal in period 1 as well. □

A useful lemma

The following Lemma is useful in the proofs of several tactical choice-related results.

Lemma 3 (Single-crossing of $\hat{c}$-comparisons under floor/ceiling and slope ordering)**.** *For $\kappa \in \{p, v\}$ define*

$$h_\kappa(x) := \int_0^1 (A_\kappa(u) - x)_+ \, du, \qquad \hat{c}_2^\kappa(x) := \int_0^1 \Pr(s \le A_\kappa(u) \mid s \sim \mathrm{Unif}[x, M]) \, du = \frac{h_\kappa(x)}{M - x},$$

where $(t)_+ := \max\{t, 0\}$.
Then there exists a unique cutoff $c \in [0, A_v(1)]$ such that for all $x \in [0, A_p(1)]$,

$$\hat{c}_2^v(x) \ge \hat{c}_2^p(x) \quad \Longleftrightarrow \quad x \le c \qquad (\textit{equivalently } h_v(x) \ge h_p(x) \Longleftrightarrow x \le c).$$

In particular, the continuation comparison between tactics is single-crossing in the posterior state x.

Proof. Since the denominator $M - x$ is common across tactics,

$$\hat{c}_2^v(x) \ge \hat{c}_2^p(x) \quad \Longleftrightarrow \quad h_v(x) \ge h_p(x).$$

Define $D(x) := h_v(x) - h_p(x)$; it suffices to show $\{x : D(x) \ge 0\}$ is an interval of the form $[0, c]$ and that c is unique.

Let $d(u) := A_p(u) - A_v(u)$. By assumptions,

$$d(0) < 0, \qquad d(1) > 0, \qquad d'(u) = A_p'(u) - A_v'(u) > 0.$$

Hence d is strictly increasing and crosses zero exactly once: there exists a unique $u^* \in (0, 1)$ such that

$$A_p(u^*) = A_v(u^*) =: x^*.$$

Moreover, for $u < u^*$ we have $A_v(u) > A_p(u)$ and for $u > u^*$ we have $A_p(u) > A_v(u)$. Because A_p, A_v are strictly increasing, their inverses are well-defined on their ranges. For every x in the overlap of ranges $[A_p(0), A_v(1)]$,

$$A_v^{-1}(x) < A_p^{-1}(x) \ \text{ if } x < x^*, \qquad A_v^{-1}(x) > A_p^{-1}(x) \ \text{ if } x > x^*.$$

We now differentiate $h_\kappa(x)$ and characterize $D'(x)$. Fix $\kappa \in \{p, v\}$ and define $u_\kappa(x) := A_\kappa^{-1}(x)$ for $x \in [A_\kappa(0), A_\kappa(1)]$. Since A_κ is strictly increasing,

$$(A_\kappa(u) - x)_+ = 0 \ \text{ for } u \le u_\kappa(x), \qquad (A_\kappa(u) - x)_+ = A_\kappa(u) - x \ \text{ for } u > u_\kappa(x).$$

Thus, for such x,

$$h_\kappa(x) = \int_{u_\kappa(x)}^1 (A_\kappa(u) - x) \, du. \tag{4}$$

Differentiate (4) using Leibniz' rule. The boundary term vanishes because $A_\kappa(u_\kappa(x)) = x$, yielding

$$h'_\kappa(x) = -\int_{u_\kappa(x)}^{1} 1\,du = -(1 - u_\kappa(x)) = -(1 - A_\kappa^{-1}(x)).$$

Hence on the overlap of ranges,

$$D'(x) = h'_v(x) - h'_p(x) = A_v^{-1}(x) - A_p^{-1}(x). \tag{5}$$

By Step 2, $D'(x) < 0$ for $x < x^*$ and $D'(x) > 0$ for $x > x^*$ (whenever both inverses are defined). Therefore D is decreasing up to x^* and increasing thereafter (i.e. D is "U-shaped").

I now show that $D(x)$ is strictly negative above $A_v(1)$. For any $x \in (A_v(1), A_p(1)]$, we have $A_v(u) \leq A_v(1) < x$ for all u, so $(A_v(u) - x)_+ \equiv 0$ and hence $h_v(x) = 0$. At the same time, since $A_p(1) > x$ and A_p is continuous and increasing, the set $\{u : A_p(u) > x\}$ has positive measure, so $h_p(x) > 0$. Thus $D(x) = h_v(x) - h_p(x) < 0$ for all $x \in (A_v(1), A_p(1)]$.

The above arguments imply that D is U-shaped on $[A_p(0), A_v(1)]$ and strictly negative for all $x \in (A_v(1), A_p(1)]$. In particular, once D becomes negative, it cannot become positive again for larger x. Therefore the set $\{x \in [0, A_p(1)] : D(x) \geq 0\}$ is either empty or an interval of the form $[0, c]$ for some $c \leq A_v(1)$. If the set is empty, define $c = 0$. Otherwise, define $c := \sup\{x : D(x) \geq 0\}$; continuity of D implies $D(c) = 0$, and the U-shape implies this c is the unique cutoff.

Finally, since $\hat{c}_2^\kappa(x) = h_\kappa(x)/(M - x)$ with common denominator, the same cutoff applies to $\hat{c}_2^v(x) \geq \hat{c}_2^p(x)$. □

Uniqueness of the game with tactical choice

Proposition 4 (Pathwise uniqueness with tactical choice)**.** *Fix a realized sequence of cost chocks $\{c_1, c_2\}$. There exists a unique Perfect Bayesian Equilibrium. In each on-path period, the continuation state can be summarized by some x s.t. $F_t = Unif[x, M]$. At each continuation, the success-maximizing social planner has a uniquely preferred tactic, and agents follow a cutoff strategy as described in Proposition 1.*

Proof. Backward induct. In period 2, Proposition 1 guarantees that for each tactic $\kappa \in \{p, v\}$ and each posterior F_2, there is a unique cutoff equilibrium with expected success probability $\hat{c}_2^\kappa$. The organizer selects $\kappa^* \in \arg\max_\kappa \hat{c}_2^\kappa$ (breaking ties in favor of peace), which is well-defined since it is a comparison of two real numbers. In period 1, each tactic τ induces a unique cutoff equilibrium (again by Proposition 1) and a well-defined continuation value W_1^τ that aggregates over the period-2 best response at each possible failure; the organizer selects the tactic maximizing W_1^τ (breaking ties in favor of peace). Since each component (cutoff participation, belief updating, and tactical choice) is uniquely pinned down at every decision node, backward induction yields a unique equilibrium. □

Proof of Lemma 2

Proof. Suppose $A_v(\ell) \equiv \bar{v}$ for all ℓ. Let $\ell^* \in (0,1)$ be the unique turnout level satisfying $A_p(\ell^*) = \bar{v}$, which exists since $A_p(0) < \bar{v} < A_p(1)$ by the floor/ceiling assumptions.

Under violence, realized power is $\bar{v}$ regardless of turnout. If this is used in the first period and then fails, the posterior becomes $\theta \mid \theta > \bar{v}$, i.e. $F_2 = \text{Unif}[\bar{v}, M]$. In the continuation, violence again produces $\bar{v}$, which is now known to be insufficient, so $\hat{c}_2^v(\bar{v}) = 0$. The organizer therefore uses peace in period 2, giving

$$\begin{aligned} W_1^v &= F_1(\bar{v}) + \big(1 - F_1(\bar{v})\big)\, \hat{c}_2^p(\bar{v}) \\ &= F_1(\bar{v}) + \int_0^1 \big(F_1(A_p(u)) - F_1(\bar{v})\big)^+ \, du, \end{aligned} \tag{6}$$

where the second line follows by expanding $\hat{c}_2^p(\bar{v}) = \int_0^1 \frac{(F_1(A_p(u)) - F_1(\bar{v}))^+}{1 - F_1(\bar{v})}\, du$ and multiplying through by $1 - F_1(\bar{v})$.

Consider a strategy in which peace is used in period 1, but the organizer commits to violence in period 2 regardless of the period-1 outcome. Its payoff is

$$\tilde{W}^{p\to v} \;=\; \int_0^1 \Big[\, F_1(A_p(\ell)) \;+\; \big(1 - F_1(A_p(\ell))\big)\, \hat{c}_2^v\big(A_p(\ell)\big) \,\Big]\, d\ell.$$

Evaluate the integrand case by case.

- If $A_p(\ell) \geq \bar{v}$ (i.e. $\ell \geq \ell^*$): the posterior after failure is $\theta \mid \theta > A_p(\ell) \geq \bar{v}$, so violence in period 2 is certain to fail, giving $\hat{c}_2^v(A_p(\ell)) = 0$. The integrand reduces to $F_1(A_p(\ell))$.
- If $A_p(\ell) < \bar{v}$ (i.e. $\ell < \ell^*$): the posterior is $\theta \mid \theta > A_p(\ell)$, and violence succeeds iff $\theta \leq \bar{v}$, so
$$\hat{c}_2^v(A_p(\ell)) \;=\; \frac{F_1(\bar{v}) - F_1(A_p(\ell))}{1 - F_1(A_p(\ell))}.$$
The integrand simplifies to $F_1(A_p(\ell)) + F_1(\bar{v}) - F_1(A_p(\ell)) \;=\; F_1(\bar{v})$.

Combining:

$$\begin{aligned} \tilde{W}^{p\to v} &= \int_0^{\ell^*} F_1(\bar{v})\, d\ell \;+\; \int_{\ell^*}^1 F_1(A_p(\ell))\, d\ell \\ &= \ell^* F_1(\bar{v}) \;+\; \int_{\ell^*}^1 F_1(A_p(\ell))\, d\ell \\ &= F_1(\bar{v}) \;+\; \int_0^1 \big(F_1(A_p(\ell)) - F_1(\bar{v})\big)^+ \, d\ell, \end{aligned} \tag{7}$$

where the last equality uses $F_1(A_p(\ell)) > F_1(\bar{v})$ iff $\ell > \ell^*$ (since A_p is strictly increasing), and splits the integral accordingly.

Comparing (6) and (7) shows that $\tilde{W}^{p\to v} = W_1^v$.

Since W_1^p allows the organizer to choose the *optimal* period-2 tactic rather than committing to violence, it must be that $W_1^p \geq \tilde{W}^{p\to v} = W_1^v$. It remains to show the inequality is strict. After a peaceful failure at power $x = A_p(\ell)$ with x slightly below $\bar{v}$, the posterior is $\theta \mid \theta > x$. Violence gives $\hat{c}_2^v(x) = \frac{F_1(\bar{v})-F_1(x)}{1-F_1(x)} \to 0$ as $x \to \bar{v}$. Peace gives

$$\hat{c}_2^p(x) = \int_0^1 \frac{\left(F_1(A_p(u)) - F_1(x)\right)^+}{1 - F_1(x)}\, du,$$

which remains bounded away from zero because $A_p(1) > \bar{v}$ ensures a positive-measure set of turnout realizations can beat regime types above $\bar{v}$. So for x in a neighborhood of $\bar{v}$, $\hat{c}_2^p(x) > \hat{c}_2^v(x)$, meaning the optimal continuation strictly improves upon the violence-only continuation.

Since this improvement holds on a positive-measure set of first-period turnout realizations (namely ℓ near ℓ^*), and each such realization is reached with positive probability conditional on failure, we conclude $W_1^p > \tilde{W}^{p\to v} = W_1^v$. $\square$.

Proof of Proposition 3

Proof. Part (i) is Lemma 2. For part (ii), define

$$H(x) := \Pr(\text{success in period 1 or 2} \mid \text{period-1 power} = x, \text{optimal period-2 play}),$$

That is, $H(x)$ describes given a realized level of power x in period 1, the total likelihood of success (either in period 1 or period 2) This is well-defined as a function of x alone because the posterior after failing at power x is Uniform$[x, M]$ regardless of which tactic generated x. Expanding,

$$H(x) = \underbrace{F_1(x)}_{\text{win in period 1}} + \underbrace{(1 - F_1(x))}_{\text{fail in period 1}} \cdot \underbrace{\max_{\kappa\in\{p,v\}} \hat{c}_2^\kappa(x)}_{\text{optimal period-2 success}}$$

where $\hat{c}_2^\kappa(x)$ is the period-2 success probability under tactic κ given posterior Uniform$[x, M]$:

$$\hat{c}_2^\kappa(x) = \int_0^1 \frac{\max\{A_\kappa(u) - x,\, 0\}}{M - x}\, du$$

Substituting into H and using $F_1(x) = x/M$, the factor $(M - x)$ cancels between the failure probability and the denominator of $\hat{c}_2^\kappa$:

$$H(x) = \frac{x}{M} + \frac{1}{M} \max_{\kappa\in\{p,v\}} \underbrace{\int_0^1 \max\{A_\kappa(u) - x,\, 0\}\, du}_{=:\, h_\kappa(x)}$$

Here $h_\kappa(x)$ is the unnormalized expected period-2 power surplus under tactic κ, i.e. the portion of κ's power range that exceeds the truncation point x. Computing explicitly:

$$h_p(x) = \int_x^1 (u - x)\, du = \frac{(1-x)^2}{2} \quad \text{and} \quad h_v(x) = \int_0^1 \max\{\alpha + \beta u - x,\, 0\}\, du.$$

Hence

$$H(x) = \frac{1}{M}\Big(x + \max\{h_p(x), h_v(x)\}\Big)$$

Step 1: H is weakly increasing. I show $H'(x) \geq 0$ in each region determined by which tactic is period-2 optimal.

- Case 1: Peace is period-2 optimal, i.e. $h_p(x) \geq h_v(x)$. Then $H(x) = \frac{1}{M}\left[x + \frac{(1-x)^2}{2}\right] = \frac{1+x^2}{2M}$, so $H'(x) = \frac{x}{M} \geq 0$.
- Case 2: Violence is period-2 optimal and $x \leq \alpha$ (so the max operator does not bind). Then, $\alpha + \beta u \geq \alpha \geq x$ for all u, so $h_v(x) = \alpha + \beta/2 - x \implies H(x) = \frac{1}{M}(\alpha - \beta/2) \implies H'(x) = 0$.
- Case 3: Violence is period-2 optimal and $x \in (\alpha, \alpha+\beta)$. Then $\max\{\alpha + \beta u - x, 0\} > 0$ iff $u > (x-\alpha)/\beta$, so

  $$h_v(x) = \int_{(x-\alpha)/\beta}^1 (\alpha + \beta u - x)\, du = \frac{(\alpha+\beta-x)^2}{2\beta},$$

  and $h_v'(x) = -\frac{\alpha+\beta-x}{\beta}$. Thus

  $$H'(x) = \frac{1}{M}\left(1 - \frac{\alpha+\beta-x}{\beta}\right) = \frac{x-\alpha}{\beta M} \geq 0,$$

  with strict inequality for $x > \alpha$.

Next we note that H is continuous and weakly increasing at the boundaries between the regions. At $x = \alpha$, both Cases 2 and 3 yield $H(\alpha) = \frac{1}{M}(\alpha + \beta/2)$. At the switching point x^* where $h_p(x^*) = h_v(x^*)$, both branches of the max operator agree, so H is continuous. Since H is continuous, weakly increasing within each region, and does not jump down at the boundaries, it is weakly increasing on $[0, \alpha+\beta]$.

Since $H'(x) \geq 0$ in every region, H is weakly increasing, and strictly increasing for $x > \alpha$.

Step 2: Connection to W_1^τ. Recall that turnout ℓ is uniform on $[0,1]$, so

$$W_1^\tau = \int_0^1 H(A_\tau(\ell))\, d\ell$$

With a change of variables so that $x = A_\tau(\ell)$:

$$W_1^p = \int_0^1 H(x;\beta)dx \quad \text{and} \quad W_1^v = \frac{1}{\beta}\int_\alpha^{\alpha+\beta} H(x;\beta)dx$$

where

$$H(x;\beta) = \frac{1}{M}\Big(x + \max\{h_p(x), h_v(x;\beta)\}\Big)$$

Step 3: Comparison at $\beta = 1-\alpha$:
At $\beta = 1-\alpha$, $A_v(\ell) = \alpha + (1-\alpha)\ell > \ell = A_p(\ell)$ for all $\ell \in (0,1)$, since $A_v(\ell) - A_p(\ell) = \alpha(1-\ell) > 0$.

To establish $H(A_v(\ell)) > H(A_p(\ell))$ for all $\ell \in (0,1)$, consider two cases.

- For $\ell > \alpha$, both $A_p(\ell)$ and $A_v(\ell)$ exceed α, and since H is strictly increasing for $x > \alpha$ (Step 1), the inequality follows directly.
- For $\ell \le \alpha$, $A_p(\ell) = \ell \le \alpha$ while $A_v(\ell) = \alpha + (1-\alpha)\ell > \alpha$ for all $\ell > 0$. Since H is constant on $[0,\alpha]$ (Case 2) and strictly increasing for $x > \alpha$, $H(A_v(\ell)) > H(\alpha) = H(A_p(\ell))$.

In both cases $H(A_v(\ell)) > H(A_p(\ell))$, so integrating yields $W_1^v > W_1^p$.

Step 4: Existence of β^*. Note that $H(x;\beta)$ is jointly continuous in (x,β): $h_p(x)$ is independent of β, $h_v(x;\beta)$ is continuous in both arguments, and the max of continuous functions is continuous. It follows that $W_1^v(\beta) = \int_0^1 H(\alpha + \beta\ell;\, \beta)\, d\ell$ and $W_1^p(\beta) = \int_0^1 H(\ell;\, \beta)\, d\ell$ are both continuous in β on $(0, 1-\alpha)$.

Define $D(\beta) := W_1^v(\beta) - W_1^p(\beta)$. By part (i), $D(\beta) < 0$ for β near zero; by Step 3, $D(\beta) > 0$ for β near $1-\alpha$. By the intermediate value theorem, D has at least one zero.

Let $\beta^*(\alpha) := \sup\{\beta \in (0, 1-\alpha) : D(\beta) \le 0\}$. Then for any $\hat{\beta} > \beta^*(\alpha)$ we must have $D(\hat{\beta}) > 0$; otherwise $\hat{\beta}$ would belong to the set $\{D \le 0\}$, contradicting $\hat{\beta} > \sup\{D \le 0\}$. Moreover, continuity implies $D(\beta^*(\alpha)) = 0$ (unless $\beta^*(\alpha) = 0$, which is ruled out because $D(1-\alpha) > 0$ while $D(0) < 0$). Finally, by continuity, $D(\beta) < 0$ in a neighborhood of zero.

Step 5: Uniqueness of β^*. Define the (rescaled) difference

$$\Delta(\beta) \;:=\; M\big(W_1^v(\beta) - W_1^p(\beta)\big),$$

which describes the *ex ante* advantage of choosing violence vs peace in period 1, taking into account both:

1. How the tactic chosen in period 1 changes the distribution of realized protest power x_1
2. The mapping from x_1 to the optimal period-2 choice and likelihood of success

(A) Rewrite payoffs as integrals over realized power. For each β, write

$$\widetilde{H}_\beta(x) \;:=\; x + \max\{h_p(x), h_v^\beta(x)\},$$

so that $H(x) = \widetilde{H}_\beta(x)/M$ in Step 1. Because $\ell \sim \text{Unif}[0,1]$ from the marginal agent's perspective, a change of variables gives

$$MW_1^p(\beta) = \int_0^1 \widetilde{H}_\beta(A_p(\ell))\, d\ell = \int_0^1 \widetilde{H}_\beta(x)\, dx,$$
$$MW_1^v(\beta) = \int_0^1 \widetilde{H}_\beta(A_v(\ell))\, d\ell = \frac{1}{\beta}\int_\alpha^{\alpha+\beta} \widetilde{H}_\beta(x)\, dx,$$

hence

$$\Delta(\beta) = \frac{1}{\beta}\int_\alpha^{\alpha+\beta} \widetilde{H}_\beta(x)\, dx \;-\; \int_0^1 \widetilde{H}_\beta(x)\, dx.$$

By construction, $\widetilde{H}_\beta$ is continuous in (x, β), so Δ is continuous in β.

(B) Closed-form expressions for h_p and h_v^β. On $x \in [0,1]$,

$$h_p(x) = \int_x^1 (u - x)\, du = \frac{(1-x)^2}{2}.$$

For violence, recall $A_v(u) = \alpha + \beta u$ and define

$$h_v^\beta(x) = \int_0^1 \max\{\alpha + \beta u - x, 0\}\, du = \begin{cases} \alpha + \beta/2 - x, & x \le \alpha, \\ \dfrac{(\alpha+\beta-x)^2}{2\beta}, & \alpha < x < \alpha+\beta, \\ 0, & x \ge \alpha+\beta. \end{cases}$$

It will be convenient to also record the two branch values of $\widetilde{H}_\beta(x) = x + \max\{h_p(x), h_v^\beta(x)\}$:

$$\widetilde{H}^p(x) := x + h_p(x) = \frac{1+x^2}{2}, \qquad \widetilde{H}^{v,\beta}(x) := x + h_v^\beta(x) = \begin{cases} \alpha + \beta/2, & x \le \alpha, \\ \alpha + \beta/2 + \dfrac{(x-\alpha)^2}{2\beta}, & \alpha < x < \alpha+\beta. \end{cases}$$

(C) A cutoff representation.

Lemma 4 (Single-crossing of period-2 tactic choice)**.** *For each $\beta \in (0, 1-\alpha)$, the difference $\tilde{H}^{v,\beta}(x) - \tilde{H}^p(x)$ crosses zero at most once on $[0,1]$. That is, there exists a unique cutoff $c(\alpha,\beta) \in [0,1]$ such that violence is (weakly) optimal in period 2 iff $x \le c(\alpha,\beta)$.*

Proof. We show that $\tilde{H}^{v,\beta}(x) - \tilde{H}^p(x)$ is eventually strictly negative and cannot return to zero once it becomes negative.

Step 1: The difference is strictly negative at $x = \alpha+\beta$ and beyond. For $x \ge \alpha+\beta$, $h_v^\beta(x) = 0$, so $\tilde{H}^{v,\beta}(x) = x$ while $\tilde{H}^p(x) = x + (1-x)^2/2 > x$. For $x = \alpha+\beta$ specifically,

$$\tilde{H}^p(\alpha+\beta) - \tilde{H}^{v,\beta}(\alpha+\beta) = \frac{(1-\alpha-\beta)^2}{2} > 0,$$

since $\alpha + \beta < 1$ by assumption.

Step 2: Shape of the difference on $[0, \alpha+\beta]$. On the flat region $x \le \alpha$:

$$\tilde{H}^{v,\beta}(x) - \tilde{H}^{p}(x) = \left(\alpha + \frac{\beta}{2}\right) - \frac{1+x^2}{2},$$

which has derivative $-x \le 0$, so the difference is weakly decreasing.

On the quadratic region $x \in (\alpha, \alpha+\beta)$:

$$\frac{d}{dx}\left[\tilde{H}^{v,\beta}(x) - \tilde{H}^{p}(x)\right] = \frac{x-\alpha}{\beta} - x = -\frac{x(1-\beta)-\alpha}{\beta},$$

which is negative for $x < \alpha/(1-\beta)$ and positive for $x > \alpha/(1-\beta)$. Thus the difference is U-shaped on this region: it continues decreasing, reaches a minimum, and then increases.

Step 3: Single crossing. The difference is (weakly) decreasing then U-shaped on $[0, \alpha+\beta]$, and strictly negative at $x = \alpha+\beta$ (Step 1). Since the increasing part of the U-shape does not return to zero by $x = \alpha+\beta$, and the difference remains strictly negative for all $x > \alpha+\beta$, the difference crosses zero at most once. If $\alpha + \beta/2 > 1/2$ (i.e., the difference is positive at $x = 0$), there is exactly one crossing, which defines $c(\alpha, \beta)$. If $\alpha + \beta/2 \le 1/2$, the difference is everywhere non-positive, and we set $c(\alpha, \beta) = 0$. □

The above Lemma implies that for each β, the comparison $\widetilde{H}^{v,\beta}(x) \gtrless \widetilde{H}^{p}(x)$ implies a unique cutoff $c(\alpha, \beta) \in [0, 1]$ such that violence is (weakly) optimal in period 2 iff $x \le c(\alpha, \beta)$, i.e.

$$\widetilde{H}_\beta(x) = \begin{cases} \widetilde{H}^{v,\beta}(x), & x \le c(\alpha,\beta), \\ \widetilde{H}^{p}(x), & x \ge c(\alpha,\beta). \end{cases}$$

Moreover, $c(\alpha, \beta)$ is explicit:

$$c(\alpha,\beta) = \begin{cases} 0, & \beta \le 1-2\alpha, \\ \sqrt{2\alpha+\beta-1}, & 1-2\alpha < \beta \le (1-\alpha)^2, \\ \dfrac{\alpha+\beta-\sqrt{\beta}}{1-\sqrt{\beta}}, & \beta > (1-\alpha)^2. \end{cases}$$

(The first two lines come from solving $\widetilde{H}^{p}(x) = \alpha + \beta/2$ on $x \le \alpha$, and the last line from solving $\widetilde{H}^{p}(x) = \alpha + \beta/2 + (x-\alpha)^2/(2\beta)$ on $(\alpha, \alpha+\beta)$.)

(D) Strict monotonicity of $\Delta(\beta)$. We show $\Delta'(\beta) > 0$ for all $\beta \in (0, 1-\alpha)$ by cases.

Case 1: $\beta \le 1-2\alpha$**.** Violence is never used in the second period.

Here $c(\alpha, \beta) = 0$ and $\widetilde{H}_\beta(x) = \widetilde{H}^{p}(x) = (1+x^2)/2$ for all $x \in [0, 1]$. Thus

$$\Delta(\beta) = \frac{1}{\beta}\int_{\alpha}^{\alpha+\beta} \frac{1+x^2}{2}\,dx - \int_0^1 \frac{1+x^2}{2}\,dx = \left(\frac{1}{2} + \frac{\alpha^2}{2} + \frac{\alpha\beta}{2} + \frac{\beta^2}{6}\right) - \frac{2}{3},$$

so

$$\Delta'(\beta) = \frac{\alpha}{2} + \frac{\beta}{3} > 0.$$

Case 2: $1 - 2\alpha < \beta \leq (1-\alpha)^2$**.** Violence is used in the second period condition upon sufficiently low realizations of first-period protest power.

Here $c(\alpha, \beta) = \sqrt{2\alpha + \beta - 1} \in (0, \alpha]$, so $\widetilde{H}_\beta(x) = \alpha + \beta/2$ on $[0, c(\alpha, \beta)]$ and $\widetilde{H}_\beta(x) = \widetilde{H}^p(x)$ on $[c(\alpha, \beta), 1]$. In particular, since $[\alpha, \alpha + \beta] \subseteq [c(\alpha, \beta), 1]$, we still have

$$\frac{1}{\beta} \int_\alpha^{\alpha+\beta} \widetilde{H}_\beta(x)\, dx = \frac{1}{\beta} \int_\alpha^{\alpha+\beta} \widetilde{H}^p(x)\, dx,$$

so the only β-dependence in the second term comes from the interval $[0, c(\alpha, \beta)]$. A direct calculation yields

$$\Delta'(\beta) = \frac{1}{6}\Big(2c(\alpha, \beta)^2 - 3c(\alpha, \beta) + 2 - \alpha\Big), \qquad c(\alpha, \beta) = \sqrt{2\alpha + \beta - 1}.$$

To see this is positive, note that in this case $c(\alpha, \beta) \in [\underline{c}, \alpha]$, where $\underline{c} := \max\{0, \sqrt{2\alpha - 1}\}$. Define $g(c) := 2c^2 - 3c + 2 - \alpha$. So, $\Delta'(\beta) = \frac{1}{6} g(c(\alpha, \beta))$. Note that g is convex in c and its minimum over $[\underline{c}, \alpha]$ occurs at either $c = 3/4$ (if $3/4 \in [\underline{c}, \alpha]$) or at an endpoint. To finish this case, it suffices to prove that $g > 0$ for all $c \in [\underline{c}, \alpha]$.

- If $\alpha \leq 3/4$, then g is decreasing on $[0, \alpha]$ and the minimum is at the rightmost end of the interval, $c = \alpha$. Evaluating: $g(\alpha) = 2\alpha^2 - 4\alpha + 2 = 2(1-\alpha)^2 > 0$.
- If $\alpha > 3/4$ and $\underline{c} \leq 3/4$, then the minimum is interior and located at $c = 3/4$, so we need to show that $g(3/4) = 7/8 - \alpha > 0$. This holds because $c \leq 3/4 \iff \alpha \leq \frac{25}{32} < \frac{7}{8}$, where $\alpha \leq \frac{25}{32}$ because $\sqrt{2\alpha - 1} \leq 3/4$.
- If $\alpha > 3/4$ and $\underline{c} \geq 3/4$, then g is increasing over the interval $[\underline{c}, \alpha]$ so the minimum is at the left endpoint $c = \underline{c} = \sqrt{2\alpha - 1}$. Hence, $g(\underline{c}) = 4\alpha - 3\sqrt{2\alpha - 1} - \alpha = 3\big(\alpha - \sqrt{2\alpha - 1}\big) > 0$, where the strict inequality follows from $\alpha < 1$ (since $\alpha > \sqrt{2\alpha - 1} \iff (\alpha - 1)^2 > 0$).

Hence $g(c(\alpha, \beta)) > 0$ and therefore $\Delta'(\beta) > 0$ throughout Case 2.

Case 3: $\beta > (1-\alpha)^2$**.** That is, after a period-1 failure, period-2 violence is optimal for all histories with sufficiently low realized power $x \leq c(\alpha, \beta)$, and $c(\alpha, \beta) \in (\alpha, \alpha + \beta)$. It is convenient to write $y := \sqrt{\beta} \in (0, 1)$, so $\beta = y^2$ and

$$c(y) = \frac{\alpha + y^2 - y}{1 - y}.$$

Integration of $\widetilde{H}_\beta$ on $[0, \alpha]$, $[\alpha, c(\alpha, \beta)]$, and $[c(\alpha, \beta), 1]$, together with the change of variable $y = \sqrt{\beta}$, yields an explicit formula for $\Delta(y)$:

$$\Delta(y) = \frac{2\alpha^3 + 3\alpha^2 y^2 + 3\alpha^2 y - 6\alpha^2 + 3\alpha y^4 + 3\alpha y^3 - 6\alpha y^2 - 6\alpha y + 6\alpha + y^6 + y^5 - 2y^4 - 4y^3 + 3y^2 + 3y - 2}{6y(y-1)}$$

Differentiating with respect to y:

$$\frac{d}{dy}\Delta(y) = \frac{\Psi(\alpha, y)}{3y^2(1-y)^2}$$

where $\Psi(\alpha, y)$ is the cubic polynomial in α:

$$\Psi(\alpha, y) = (1-2y)\alpha^3 - 3(1-y)^2\alpha^2 + 3(1-y)^2(y^3+y^2+1)\alpha + (1-y)^2(2y^5+3y^4-2y^2-1).$$

Since the denominator $3y^2(1-y)^2$ is strictly positive, we just need to show $\Psi(\alpha, y) > 0$ on the domain and we are done.

In Case 3, $\beta > (1-\alpha)^2$ and $\beta < 1-\alpha$ are equivalent to $1-y < \alpha < 1-y^2$. Fix any $y \in (0,1)$ and consider $\Psi(\cdot, y)$ as a function of α on the interval $[1-y,\, 1-y^2]$. First, plugging in each of the endpoints, we can see that they evaluate to be positive:

$$\Psi(1-y, y) = 2y^5(1-y)^2 > 0$$
$$\Psi(1-y^2, y) = y^3(1-y)^2(y^2+1) > 0$$

Second, the derivative $\partial_\alpha \Psi(\alpha, y)$ is a quadratic in α, and evaluating it at the endpoints gives:

$$\partial_\alpha \Psi(1-y, y) = 3y^2(1-y)^2(y+1) > 0, \qquad \partial_\alpha \Psi(1-y^2, y) = -3y^3(1-y)^2 < 0.$$

Because a quadratic cannot change sign twice on an interval when it has opposite signs at the endpoints, $\partial_\alpha \Psi(\alpha, y)$ must have only one root in $(1-y,\, 1-y^2)$. Hence $\Psi(\cdot, y)$ is increasing and then decreasing on $[1-y,\, 1-y^2]$, so its minimum on this interval is attained at an endpoint. Since both endpoint values are strictly positive, $\Psi(\alpha, y) > 0$ for all $\alpha \in (1-y,\, 1-y^2)$. Therefore $\frac{d}{dy}\Delta(y^2) > 0$, and since $\frac{d}{d\beta}\Delta(\beta) = \frac{1}{2\sqrt{\beta}}\frac{d}{dy}\Delta(y^2)$, we conclude $\Delta'(\beta) > 0$ throughout Case 3 as well.

Combining the three cases, $\Delta(\beta)$ is continuous and strictly increasing on $(0, 1-\alpha)$. From Lemma 2, $\Delta(0) < 0$, and part (ii), $\Delta(1-\alpha) > 0$. By the intermediate value theorem and strict monotonicity, there exists a *unique* $\beta^* \in (0, 1-\alpha)$ such that $\Delta(\beta^*) = 0$, i.e. $W_1^v(\beta^*) = W_1^p(\beta^*)$, and moreover $W_1^v > W_1^p$ iff $\beta > \beta^*$.

Finally, it remains to be shown that $\beta^*(\alpha) > (1-\alpha)^2$ (or equivalently, whenever the success-maximizing organizer leads with violence, the continuation problem lies in the region $c(\alpha, \beta) > \alpha$.) By the previous proof, $\Delta(\alpha, \beta)$ is strictly increasing in β on $(0, 1-\alpha)$ and has a unique zero at $\beta^*(\alpha)$, meaning that the statement $\beta^*(\alpha) > (1-\alpha)^2$ is equivalent to $\Delta(\alpha, (1-\alpha)^2) < 0$.

At $\beta = (1-\alpha)^2$ we are at the boundary of Case 2, where $c(\alpha, \beta) = \alpha$ and $\widetilde{H}_\beta(x) = \widetilde{H}^p(x)$ for all $x \in [\alpha, \alpha+\beta]$. Using the Case 2 closed form derived in the proof of Proposition 3,

$$\Delta(\alpha, \beta) = \left(\frac{1}{2} + \frac{\alpha^2}{2} + \frac{\alpha\beta}{2} + \frac{\beta^2}{6}\right) - \left(\frac{2}{3} + \frac{c(\alpha, \beta)^3}{3}\right).$$

Substituting $\beta = (1-\alpha)^2$ and $c(\alpha, \beta) = \alpha$ yields

$$\Delta(\alpha, (1-\alpha)^2) = \left(\frac{1}{2} + \frac{\alpha^2}{2} + \frac{\alpha(1-\alpha)^2}{2} + \frac{(1-\alpha)^4}{6}\right) - \left(\frac{2}{3} + \frac{\alpha^3}{3}\right) = -\frac{\alpha(1-\alpha)^3}{6} < 0.$$

Since $\Delta(\alpha, \beta)$ is strictly increasing in β, it follows that the unique root satisfies $\beta^*(\alpha) > (1-\alpha)^2$. $\square$